

Lab 6: Frames, Logic & Offsets

EE0849 Introduction to Robotics

Basri Erdogan

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Safety Reminder

- Start in **T1 mode** — max 250 mm/s
- Hold the **Deadman Switch** at all times
- Always test new programs in **single-step** mode first
- Know the **E-Stop** locations before starting

Prerequisites

This lab builds on Lab 5 (FANUC Hardware Introduction). You should already be comfortable with jogging, teaching positions, and running simple J/L motion programs.

0.1 Teach Pendant Quick Reference

Locate the key buttons on the teach pendant before starting:

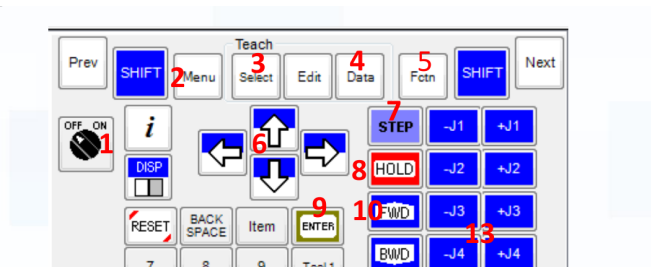


Figure 1: Teach pendant button layout

Key buttons: 1=Enable switch, 2=Menu, 3=Select, 4=Data, 5=Fctn, 6=Cursor, 7=Step, 8=Hold, 9=Enter, 10=FWD, 11=Coord, 12=Override +/-, 13=Jog keys

Controller front panel (locate before starting): **E-Stop** (red mushroom button, top right), **Three-Mode Switch** (T1/T2/AUTO, left side), **Cycle Start** (top left), **Power switch** (lower left).

0.2 Objectives

In this lab you will:

1. Set up a **Tool Frame** (UTOOL) to define the end-effector TCP
2. Create a **User Frame** (UFRAME) to define a work surface coordinate system
3. Use **Position Registers** (PR) and offsets for flexible positioning
4. Write programs with **IF/ELSE** conditional logic and **FOR** loops
5. Combine these into a repeating pattern task using offsets

1 Part 1: Tool Frame Setup (UTOOL)

1.1 What Is a Tool Frame?

Why Define a Tool Frame?

The robot's default TCP is at the center of the tool flange. When you attach a gripper or any tool, the actual working point is offset from the flange. A **Tool Frame** (UTOOL) tells the controller where the real TCP is, so that:

- **WORLD** jogging moves the tool tip, not the flange
- Cartesian positions (L motion) are accurate
- Tool rotations happen about the correct point

1.2 Jogging the Robot

To jog the robot for teaching points:

1. Set the mode switch to **T1**
2. Enable the teach pendant (flip the **Enable switch**)
3. Hold the **Deadman switch** (back of pendant) in the **middle** position
4. Hold **SHIFT** + press a **Jog key** (+J1/-J1, +J2/-J2, etc.)
5. Press **COORD** to switch between JOINT, WORLD, and TOOL coordinate systems

1.3 Direct Entry Method

The TCP is simply a measured offset from the tool flange to the actual working point (e.g., the gripper jaw tips). If you know the gripper dimensions, you can enter the offset directly — no special calibration procedure needed.

Exercise: Measure and Enter TCP Offset

1. Using calipers or a ruler, measure the distance from the **flange face** to the **grripper jaw tips** (gripping point)
2. This is your **Z offset** (typically the largest value)
3. If the gripper is centered on the flange, X and Y offsets are **0**
4. If the gripper is mounted straight (not angled), W, P, R are **0**
5. Enter the values on the teach pendant (see below)

Teach Pendant

MENU → **6 SETUP** → **3 Frames** → **F3 [OTHER]** → select **Tool Frame**

A list of 10 frames appears (UTOOL[1]–UTOOL[10]). Select an empty slot (e.g., UTOOL[2]).

F2 [METHOD] → **Direct Entry**

Enter the measured values: X = 0, Y = 0, Z = _____ mm, W = 0, P = 0, R = 0

Record your measurements:

Measurement	Value (mm)
Flange face to jaw tips (Z)	
Lateral offset (X)	
Lateral offset (Y)	

Why Not Three-Point Method?

The three-point method requires touching a sharp reference point from three different orientations. This works well for tools with a pointy tip (welding torch, deburring tool), but is difficult with a gripper — there is no single sharp point to touch. Direct entry is more practical for grippers when the dimensions are known.

Exercise: Verify Tool Frame

1. Activate your new UTOOL: Setup → Frames → Tool Frame → select your frame from the list
2. Switch to **WORLD** jog mode and jog in Z only
3. The jaw tips should move straight up/down. Does it? _____
4. Rotate W/P/R — the jaw tips should stay roughly in place. Does it? _____

2 Part 2: User Frame Setup (UFRAME)

2.1 What Is a User Frame?

Why Define a User Frame?

A **User Frame** (UFRAME) defines a coordinate system on your work surface. Benefits:

- Positions are relative to the **work surface**, not the robot base
- If you move the workstation, re-teach only the UFRAME — all positions update
- Easier to reason about X/Y/Z when they align with the workpiece

2.2 Three-Point Method

You teach three points on the work surface to define the origin, X direction, and Y direction.

Exercise: Teach User Frame

1. Identify a flat work surface (e.g., the table or a fixture plate)
2. Choose an **origin** point (corner of the work area)
3. Choose a point along the **+X direction** (along one edge)
4. Choose a point along the **+Y direction** (along the perpendicular edge)
5. Record each point using the pendant

Teach Pendant

MENU → **6 SETUP** → **3 Frames** → **F3 [OTHER]** → select **User Frame**

A list of 10 frames appears (UFRAME[1]–UFRAME[10]). Select an empty slot (e.g., UFRAME[2]).

F2 [METHOD] → **Three Point**

Jog to each point, then **SHIFT + F5 [RECORD]**:

Point 1: Origin Point 2: +X direction Point 3: +Y direction

Exercise: Verify User Frame

1. Activate your new UFRAME: Setup → Frames → User Frame → select your frame from the list
2. The jog display shows the current position (press **COORD** to toggle JOINT/WORLD view)
3. Jog the robot to your origin point — X, Y, Z should read approximately 0, 0, 0
4. Jog along the table edge — only X should change. Does it? _____

Frame Warning

All taught positions are stored relative to the **active** UTOOL and UFRAME at the time of teaching. If you change frames after teaching positions, those positions will be wrong. Always set your frames **before** teaching any points.

2.3 Programming Essentials — Visual Reference

Creating a program: Press **SELECT**, then **F2 (CREATE)**, type a name, press **ENTER**.

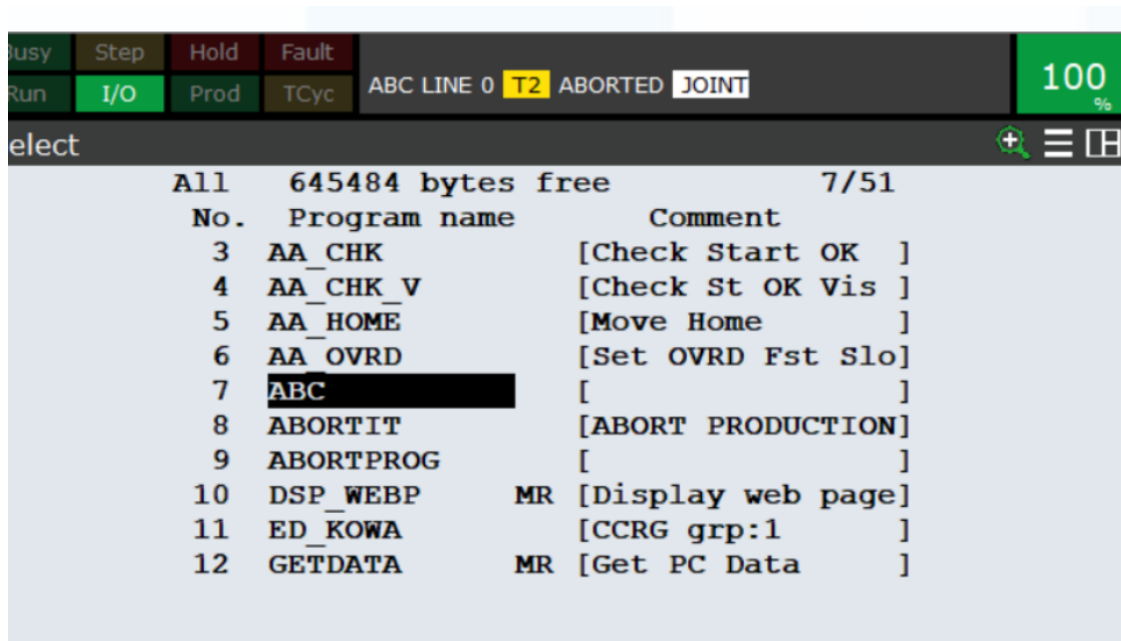


Figure 2: SELECT screen — press F2 CREATE

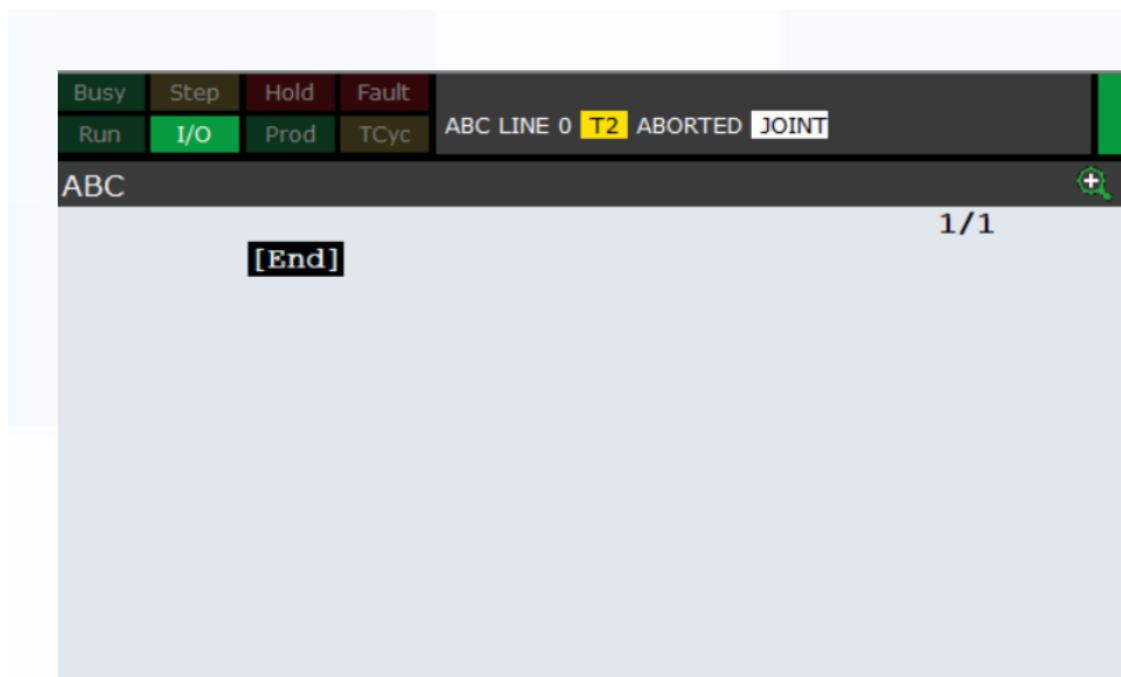


Figure 3: Empty program after creation

Recording positions: Jog the robot to the desired pose, then press **SHIFT+POINT** (defaults to Joint motion) or **POINT** alone to choose the motion type:



Figure 4: Motion type selection — J (joint), L (linear), FINE or CNT

Before running: Set override to **10%** and enable **STEP** mode (one line at a time):



Figure 5: Override set to 10 (green circle, top right)

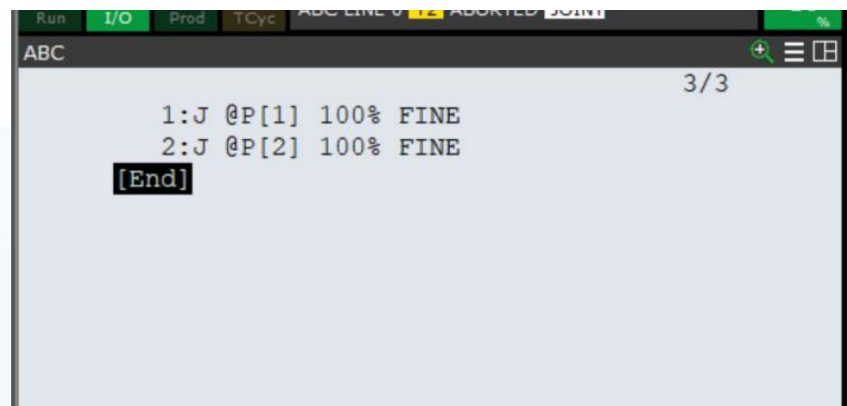


Figure 6: Program with two taught positions

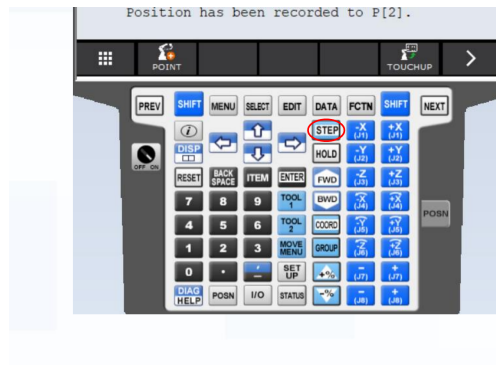


Figure 7: Pendant with STEP button circled (red) — press to toggle step mode

△ First Run Checklist

1. Set override to **10%** (use the +/- buttons)
2. Enable **STEP** mode (press STEP button — check status bar)
3. Hold **Deadman + SHIFT**, then press **FWD** to execute one line
4. Only after a successful full run: disable STEP and gradually increase override

3 Part 3: Registers and Position Offsets

3.1 Numeric Registers (R[]) and Position Registers (PR[])

Registers

- **R[n]** — stores a single number (counter, distance, flag)
- **PR[n]** — stores a full 6-DOF pose (X, Y, Z, W, P, R)
- Both can be read, written, and used in math expressions on the pendant

Teach Pendant

View/edit registers:

DATA → **F1 [TYPE]** → choose **Registers** or **Position Reg**

In a program, assign a register:

[INST] → **1 Registers** (for R[], PR[], SR[])

3.2 Using Offsets

You can offset a taught position by a Position Register. This lets you reuse one taught point for multiple locations.

Exercise: Offset Motion

1. Teach a reference point **P[1]** on the work surface
2. On the pendant, go to **DATA** → Position Reg and set **PR[1]** to:
 - X = 50, Y = 0, Z = 0, W = 0, P = 0, R = 0
3. Create a test program:

```
1: L P[1] 200mm/sec FINE ;           ! Go to reference
2: L P[1] 200mm/sec FINE Offset,PR[1] ; ! 50mm in +X
```

Teach Pendant

To add an offset to a motion line:

Cursor to the end of the line → **F4 [CHOICE]** → **Offset,PR[]**

Enter the PR number.

- Does the robot move exactly 50mm in X from P[1]? _____
- Change PR[1] to X=0, Y=50, Z=0. Run again. What happens? _____

4 Part 4: Conditional Logic – IF and FOR

4.1 IF Statements

IF on FANUC

FANUC TP does not have a true IF/ELSE block. Instead, you use **IF condition, JMP LBL[n]** to skip to a label. Structure your logic as:

1. Test the condition — if true, jump past the “else” code
2. The lines immediately after the IF are the “else” (false) path
3. The label marks where the “true” path continues

Teach Pendant

[INST] → 3 IF/SELECT → 1 IF R[] or 2 IF DI[]

Fill in the condition, then select: **JMP LBL[n]**

Exercise: Alternate Placement with IF

Write a program that places 4 objects. The first 2 go to **P[4]** (left bin) and the next 2 go to **P[5]** (right bin), decided by checking the counter register.

```

1: R[1:Count]=0 ;
2: J P[1:Home] 100% FINE ;
3: LBL[1] ;
4:   R[1:Count]=R[1:Count]+1 ;
5:   ! Pick sequence (same for all) ;
6:   L P[2:Pick_App] 500mm/sec CNT50 ;
7:   L P[3:Pick] 100mm/sec FINE ;
8:   D0[1:Gripper]=ON ;
9:   WAIT .30(sec) ;
10:  L P[2:Pick_App] 500mm/sec FINE ;
11:  ! Decide placement ;
12:  IF R[1:Count]>2, JMP LBL[20] ;
13:  ! Count <= 2: place left ;
14:  L P[4:Left] 100mm/sec FINE ;
15:  JMP LBL[30] ;
16:  LBL[20] ;
17:  ! Count > 2: place right ;
18:  L P[5:Right] 100mm/sec FINE ;
19:  LBL[30] ;
20:  D0[1:Gripper]=OFF ;
21:  WAIT .30(sec) ;
22:  IF R[1:Count]<4, JMP LBL[1] ;
23: J P[1:Home] 100% FINE ;

```

How It Works

Lines 12–18 form an IF/ELSE pattern: test the condition, jump past the “else” block if true. The false path (lines 13–15) runs when $R[1] \leq 2$, the true path (lines 17–18) runs when $R[1] > 2$. Both paths rejoin at LBL[30].

4.2 FOR Loops

Teach Pendant

[INST] → **page 3** → **1 FOR/ENDFOR** → **1 FOR R[]= ... TO ...**

Add loop body instructions, then:

[INST] → **page 3** → **1 FOR/ENDFOR** → **2 ENDFOR**

Exercise: Repeat Motion with FOR

Write a program that moves the robot to P[1], then makes 3 small “nod” motions (up 20mm, back down) using a FOR loop.

```
1: PR[1]=LP0S ;                ! Store current position
2: PR[2,3:Z offset]=20 ;       ! 20mm Z offset
3: L P[1] 200mm/sec FINE ;
4: FOR R[1:i]=1 TO 3 ;
5:   L P[1] 200mm/sec FINE Offset,PR[2] ; ! Up
6:   L P[1] 200mm/sec FINE ;           ! Back
7: ENDFOR ;
```

5 Part 5: Putting It Together – Row Pattern with Offsets

5.1 The Task

Place objects in a row of 3 positions, spaced 60mm apart in X. Instead of teaching 3 separate place points, teach **one** and use a computed offset.

Best Practice: Set Frames in the Program

Always set UT00L_NUM and UFRAME_NUM at the top of your program. This ensures the program uses the correct frames regardless of what was left active on the pendant.

Exercise: Row Placement

1. Teach: **P[1]** = Home, **P[2]** = Pick approach, **P[3]** = Pick, **P[4]** = Place approach, **P[5]** = First place position
2. Set **PR[1]** = (60, 0, 0, 0, 0, 0) — the spacing between place positions
3. Set **PR[10]** = (0, 0, 0, 0, 0, 0) — this will be the accumulated offset
4. Enter the program below and test in single-step mode

```

1: ! Row placement – 3 positions ;
2: UT00L_NUM=2 ;
3: UFRAME_NUM=2 ;
4: D0[1:Gripper]=OFF ;
5: PR[10:Offset]=(0,0,0,0,0,0) ;      ! Reset offset
4: J P[1:Home] 100% FINE ;
5: FOR R[1:i]=1 TO 3 ;
6:   ! Pick ;
7:   L P[2:Pick_App] 500mm/sec CNT50 ;
8:   L P[3:Pick] 100mm/sec FINE ;
9:   D0[1:Gripper]=ON ;
10:  WAIT .30(sec) ;
11:  L P[2:Pick_App] 500mm/sec FINE ;
12:  ! Place with offset ;
13:  L P[4:Place_App] 500mm/sec CNT50 Offset,PR[10] ;
14:  L P[5:Place] 100mm/sec FINE Offset,PR[10] ;
15:  D0[1:Gripper]=OFF ;
16:  WAIT .30(sec) ;
17:  L P[4:Place_App] 500mm/sec FINE Offset,PR[10] ;
18:  ! Increment offset for next position ;
19:  PR[10:Offset]=PR[10:Offset]+PR[1:Spacing] ;
20: ENDFOR ;
21: J P[1:Home] 100% FINE ;

```

5.2 Observations

1. Where does the robot place the 1st, 2nd, and 3rd object?
2. What would you change to place in a 2D grid (3 columns × 2 rows)?
3. What happens if you change the FOR count to 5 but only have 3 objects?

6 Quick Reference

6.1 Frame Setup

Frame	What it defines	How to access
UTOOL	TCP location relative to flange	SETUP ▢ Frames ▢ Tool Frame ▢ select from list
UFRAME	Work surface coordinate system	SETUP ▢ Frames ▢ User Frame ▢ select from list

6.2 Register Types

Register	Stores	Example
R[n]	Single number	R[1:Count]=0
PR[n]	6-DOF pose (X,Y,Z,W,P,R)	PR[1]=(60,0,0,0,0,0)
PR[n,k]	Single component of PR	PR[2,3:Z]=20 (Z component)

6.3 Logic Instructions

Instruction	Pendant path	Example
IF condition	[INST] ▢ 3 IF/SELECT	IF R[1]>2, JMP LBL[20]
FOR loop	[INST] ▢ page 3 ▢ 1 FOR/ENDFOR	FOR R[1]=1 TO 5
Label	[INST] ▢ 5 JMP/LBL ▢ 2	LBL[10]
Jump	[INST] ▢ 5 JMP/LBL ▢ 1	JMP LBL[10]
Offset	F4 on motion line	L P[1] ... Offset,PR[1]

6.4 Lab Completion

- ☐ Set up a Tool Frame using three-point method (Part 1)
- ☐ Set up a User Frame using three-point method (Part 2)
- ☐ Tested position offsets with PR registers (Part 3)
- ☐ Used IF conditions and FOR loops (Part 4)
- ☐ Completed row placement program with offsets (Part 5)

Instructor Signature: _____

Date: _____